



Fib series-iterative & recur

FIB SERIES-ITERATIVE & RECUR

Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

Sample Input:

Enter N: 7

Sample Output:

Iterative: 0 1 1 2 3 5 8

Recursive: 0 1 1 2 3 5 8

Recursive Calls: 13

Iterative Time: 0.001 ms

Recursive Time: 0.005 ms

Test Case:

Enter N: 1

Output:

0

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <time.h>
3
4 int calls = 0;
5
6 int fib(int n) {
7     calls++;
8     if (n <= 1)
9         return n;
10    return fib(n - 1) + fib(n - 2);
11 }
12
13 int main() {
14     int n, i, a = 0, b = 1, c;
15     clock_t s1, e1, s2, e2;
16
17     printf("Enter N: ");
18     scanf("%d", &n);
19
20     if (n <= 0) return 0;
21 }
```

OUTPUT

```
Enter N: Iterative: 0 1 1 2 3 5 8
Recursive: 0 1 1 2 3 5 8
Recursive Calls: 59
Iterative Time: 0.000 ms
Recursive Time: 0.000 ms
```

INPUT

7